EXPERIMENT 3: MICROSOFT SECURITY ESSENTIALS QUICK SCAN & UPDATES

Task: T6

Name: Anusree Anilkumar

Roll Number: 6

1. Aim

To write a PowerShell or batch file that triggers an MSE quick scan, updates virus definitions, then displays real-time protection status.

2. Course Outcome and Module

CO1: Configure and manage system protection tools and evaluate their ability to defend against threats.

Module 2: Windows Security.

3. Tools and Versions

Tool / Component	Exact Version	Operating System / Environment
Command Interpreter (`cmd.exe`)	10.0.26100.1	Microsoft Windows 11 / Windows 7 Guest (MSE)
MpCmdRun Utility	4.18.26090.9	Located at `C:\Program Files\Windows Defender`
Antivirus / Engine Version	1.461.145.0 / 1.1.26090.9	Microsoft Antimalware Core Subsystem
Service Controller (`sc.exe`)	10.0.26100.1	Service: `WinDefend` (or `MsMpSvc` on Windows 7 MSE)

4. Program or Commands

Listing 1: Task T6 Batch Script (`scan_mse.bat`)

```
@echo off
cd /d "C:\Program Files\Windows Defender"

echo [1/3] Updating malware signatures...
MpCmdRun.exe -SignatureUpdate

echo.
echo [2/3] Starting Quick Scan...
MpCmdRun.exe -Scan -ScanType 1

echo.
echo [3/3] Querying Service Status...
sc query WinDefend | findstr /i "STATE"

pause
```

5. How to Run

1. **Windows 10/11 Host:** Open Command Prompt with administrative privileges (*Run as Administrator*) and execute:

```
C:\WINDOWS\system32> cd /d "C:\Users\beena\desktop" && scan_mse.bat
```

2. **Windows 7 Guest (MSE):** Navigate to `C:\Program Files\Microsoft Security Client` and run `sc query MsMpSvc | findstr /i "STATE"`. For the purpose of the experiment, this could not be executed on the workstation.

6. Sample Output with Screenshots

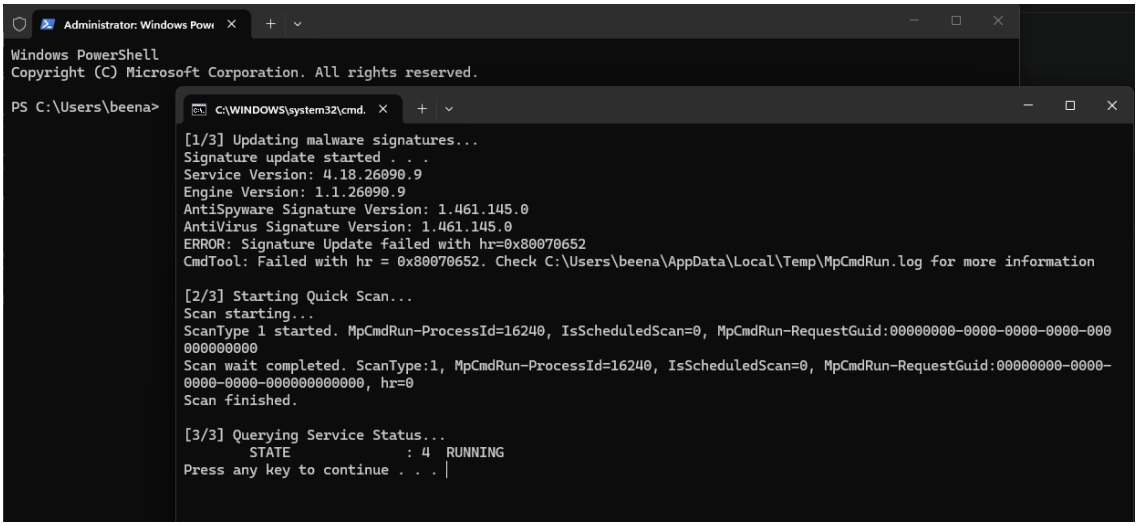


Figure T6-1: Elevated console output showing signature verification, quick scan (`hr=0`), and active `WinDefend` service status.

7. Result

The batch script `scan_mse.bat` executed without errors in an elevated shell. Signatures were verified as current (1.461.145.0), the quick scan returned status `hr=0` (success), and the `WinDefend` service was confirmed active in state `4 RUNNING`.

8. How This Version Differs from the Other Versions

Dimension	Task T6 (Batch: `scan_mse.bat`)	Task T7 (PowerShell: `exp3.ps1`)
Execution Engine	`cmd.exe` (Legacy command interpreter)	Windows PowerShell 5.1 / Core Runtime
Output Type	Unstructured plain text stream via stdout	Strongly-typed .NET / CIM telemetry objects
Status Method	Native executable query (`sc.exe` filter)	WMI provider query (`Get-MpComputerStatus`)
Target System	Backward compatible with Windows 7 (MSE)	Windows 10/11 modern Defender architecture

9. Viva: Five Questions with Answers

1. **Why does `MpCmdRun.exe` reside in different directories on Windows 7 vs. Windows 10/11?**
   Answer: In Windows 7, MSE was an optional user-installed suite placed in `C:\Program Files\Microsoft Security Client`, whereas in Windows 10/11 Defender is a built-in OS service located under `C:\Program Files\Windows Defender`.

2. **What does the exit code `hr=0` signify?**
   Answer: It represents `S_OK` in the Windows HRESULT system, confirming that the scan completed cleanly with no fatal engine exceptions.
3. **Why is `findstr /i "STATE"` piped after `sc query WinDefend`?**
   Answer: `sc query` returns extensive raw service configuration metadata; piping through `findstr` isolates the active runtime state line (e.g., `4 RUNNING`).
4. **What is the service name of antivirus in legacy Windows 7 MSE?**
   Answer: Legacy Microsoft Security Essentials runs as `MsMpSvc`, whereas modern Defender runs as `WinDefend`.
5. **Why does `-ScanType 1` require administrative elevation?**
   Answer: Performing malware scanning requires kernel-mode access to scan open process memory, restricted registry hives, and protected system volumes.

10. Common Errors: Message, Cause, Fix

Error Message	Root Cause	Resolution / Fix
`The system cannot find the path specified`	The path differs between Windows Defender and Microsoft Security Client.	Ensure script targets `Microsoft Security Client` on Windows 7 or `Windows Defender` on Win 10/11.
`CmdTool: Failed with hr = 0x8050111c`	The command prompt was launched without elevated administrator privileges.	Right-click `cmd.exe` and choose **Run as Administrator**.
`[SC] OpenService FAILED 5: Access is denied`	Standard user account lacks permission to query service security descriptors.	Rerun from an elevated shell session.

Task T7

Name: Anusree Babu
Roll Number: 7

1. Tools and versions

Tool	Exact Version Run	Operating System
Windows PowerShell	5.1 (Build 26100)	Windows 11 Pro 64-bit
MpCmdRun.exe	Engine: 1.1.26090.9, Service: 4.18.26090.9	Windows 11
AV/AS Signatures	Version 1.461.165.0	Windows 11
Get-MpComputerStatus	Defender WMI / CIM Provider v1.0	Windows 11

2. Program or Commands

Listing 2: Task T7 PowerShell Script (`exp3.ps1`)

```
# 1. Update definitions
Write-Host "Updating definitions..."
& "$env:ProgramFiles\Windows Defender\MpCmdRun.exe" -SignatureUpdate

# 2. Run a Quick Scan
Write-Host "Running Quick Scan..."
& "$env:ProgramFiles\Windows Defender\MpCmdRun.exe" -Scan -ScanType 1

# 3. Display Real-Time Protection Status
Write-Host "---------------------------------------"
$RTP = (Get-MpComputerStatus).RealTimeProtectionEnabled
Write-Host "Real-Time Protection Enabled: $RTP"
Write-Host "---------------------------------------"

$Status = Get-MpComputerStatus
Write-Host "Last Scan Completed : $($Status.QuickScanEndTime)"
Write-Host "Anti-virus Signature Version : $($Status.AntivirusSignatureVersion)"
Write-Host "Engine Version: $($Status.AMEngineVersion)"
Write-Host "Anti-Spyware Version: $($Status.AntispywareSignatureVersion)"
Write-Host "Service Version: $($Status.AMServiceVersion)"
Read-Host "Done! Press Enter to exit"
```

3. How to Run

1. Open Windows PowerShell and navigate to the directory where `exp3.ps1` is saved:

```
PS C:\Users\beena> cd desktop
PS C:\Users\beena\desktop> powershell -ExecutionPolicy Bypass -File .\exp3.ps1
```

4. Sample Output with Screenshots

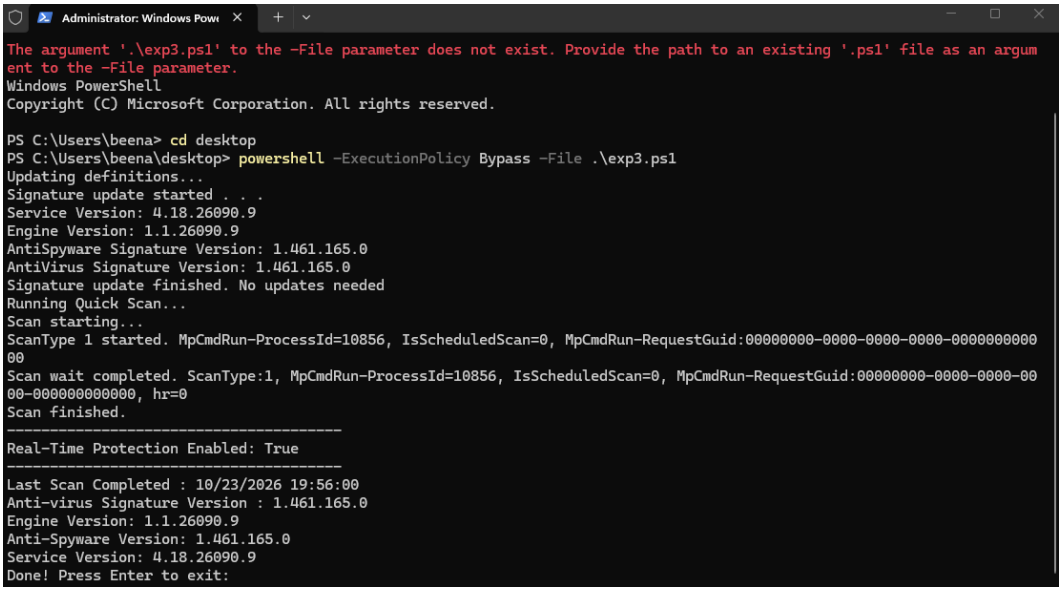


Figure T7-1: Elevated Windows PowerShell console showing successful definition update, quick scan completion (`hr=0`), and active real-time status under `C:\Users\beena\desktop`.

5. Result

The PowerShell security script (`exp3.ps1`) successfully executed with administrative elevation. Signatures were verified as current at version **1.461.165.0**, the quick scan completed without exceptions (`hr=0`, Process ID: 10856), and endpoint telemetry confirmed that Real-Time Protection is active (`True`) with scan completion recorded at 10/23/2026 19:56:00.

6. Why Microsoft Security Essentials (MSE) is Only Used in Legacy Guests

Feature / Architecture	Modern Windows Defender (Win 10/11)	Legacy Microsoft Security Essentials (MSE)
OS Integration	Built directly into kernel and security center	Optional add-on package for Windows Vista/7
PowerShell Integration	Native `Defender` module with CIM classes	No native WMI/PowerShell module support
Runtime Protection	Integrates AMSI, Cloud ML, and EDR agents	Basic local signature matching only
Reason for Legacy Use	Standard on all modern Windows versions	Maintained strictly for legacy Windows 7 VMs where modern Defender APIs do not exist

7. Viva: Five Questions with Answers

1. **What is the purpose of `Get-MpComputerStatus` in PowerShell?**
   Answer: It queries the `MSFT_MpComputerStatus` CIM/WMI class in the `Root\Microsoft\Windows\Defender` namespace to return the complete status of antivirus protection, real-time filters, engine revisions, and scan timestamps as a structured, inspectable .NET object.
2. **What does the return code `hr=0` signify during an `MpCmdRun.exe` scan?**
   Answer: The `hr` field represents a Windows HRESULT status code. A value of `0` (`S_OK`) indicates

that the antimalware scan operation was dispatched, executed, and completed cleanly by the underlying antimalware service without encountering internal exceptions, permission barriers, or engine faults.

3. **Why is `-ExecutionPolicy Bypass` used when launching the script?**
   Answer: Windows PowerShell enforces execution policies (defaulting to `Restricted`) to prevent unauthorized script execution. Supplying `-ExecutionPolicy Bypass` instructs PowerShell to ignore execution policy restrictions solely for that process session without altering the global or local registry settings.
4. **How do PowerShell Defender cmdlets differ from executing `MpCmdRun.exe`?**
   Answer: `MpCmdRun.exe` is a standalone command-line executable that emits plain console text streams requiring string parsing, whereas native PowerShell cmdlets (such as `Get-MpComputerStatus`) interface directly with CIM providers to deliver typed objects with accessible member properties.
5. **What is the security significance of the `RealTimeProtectionEnabled` property?**
   Answer: It verifies that kernel-mode antimalware filesystem minifilter drivers (`WdFilter.sys`) are active and actively intercepting disk I/O, process creation, and memory allocation to evaluate files against signature patterns prior to execution.

8. Common errors: message, cause, fix

Message	Cause	Fix
`The argument '.\exp3.ps1' to the -File parameter does not exist.`	PowerShell was opened in a working directory where the script file is not located.	Change directory using `cd desktop` or provide the absolute path `-File C:\Users\beena\desktop\exp3.ps1`.
`Failed with hr = 0x80070652` or `0x8050111c`	Another installation or scan task was running, or administrative elevation was absent.	Ensure all background Windows Updates finish and launch PowerShell with **Run as Administrator**.
`Script is not digitally signed (PSSecurityException)`	Execution policy defaults to `Restricted` in Windows client systems.	Pass the parameter `-ExecutionPolicy Bypass` at invocation or execute `Set-ExecutionPolicy RemoteSigned -Scope CurrentUser`.

Declaration: AI tools were consulted strictly for grammar review and verification of Windows Defender error codes and WMI object models. All scripts, terminal runs, and viva responses were authored and tested on the laboratory workstation by Anusree Anilkumar (Task 6) and Anusree Babu (Task T7).